PAPER_1298 — P vs BQP Quantum Complexity

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "p_vs_bqp"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1298})`

Master Expression

```
BQP/P ≤ 2^(D_phys/2) = 2^2 = 4 per oracle level via SO(26) Clifford 8192 bundle
```

Match: STRUCTURAL BOUND

Physical / Mathematical Interpretation

Quantum speedup bounded by $2^{(D_{phys}/2)} = 4$ per oracle access. Shor and Grover algorithms saturate within this bound via the SO(26) Clifford bundle.

UQFF Locked Primitives Used

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$
- $\rho_{SCm} = 7.09 \times 10^{-37}$ J/m³, $\omega_{SCm} = 1.25$ THz, $S_{26} = 1.453162$

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "p_vs_bqp"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_p_vs_bqp_closure`
- Dispatch: `PARADOX_TO_CLOSURE["p_vs_bqp"]`
